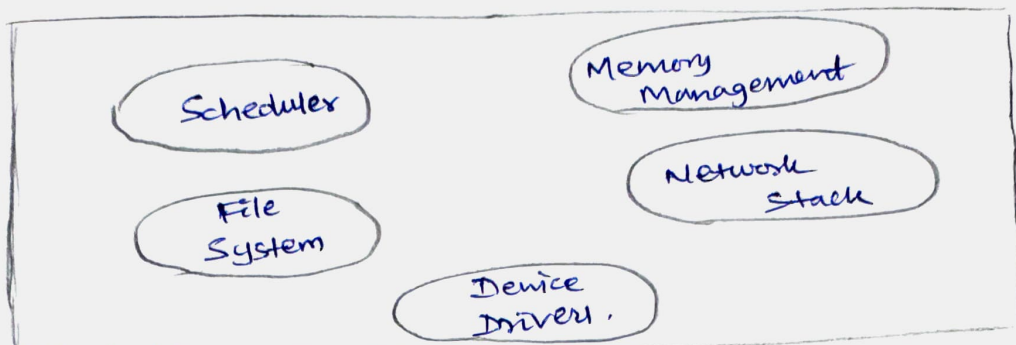




1) Scheduler

- controls how tasks are assigned to the CPU.
- Manages multitasking by deciding which process runs next, ensuring fair, CPU time using priorities & time slices.

2) Memory Management

- Control system memory resources.
- Handles allocation or de-allocation, virtual memory, paging and ensures memory protection b/w processes.

3) File System

- Manages data storage & retrieval.
- Supports file systems, handles data blocks, metadata and file-related system calls like open, read, write.

4) Network Stack

- Manages network operations & communication.
- Runs protocols like TCP/IP, handles packet transmission and ensures reliable data delivery.

5) Device Drivers

- Interface b/w hardware & software.
- Abstracts hardware details, allowing the kernel & apps to use standard APIs to communicate with devices.